Počítačové cvičení č. 6

Zvýrazňování 3

Lokální 2D operátory – zostřování

Číslicové zpracování obrazů

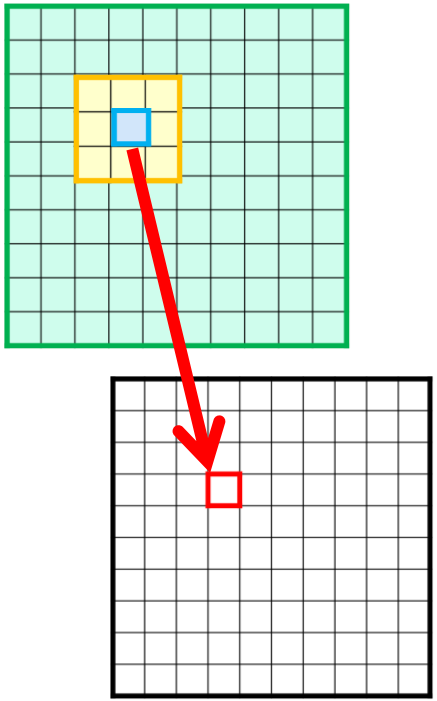
Petra Nemčeková
Ústav biomedicínského inženýrství

Typy operátorů

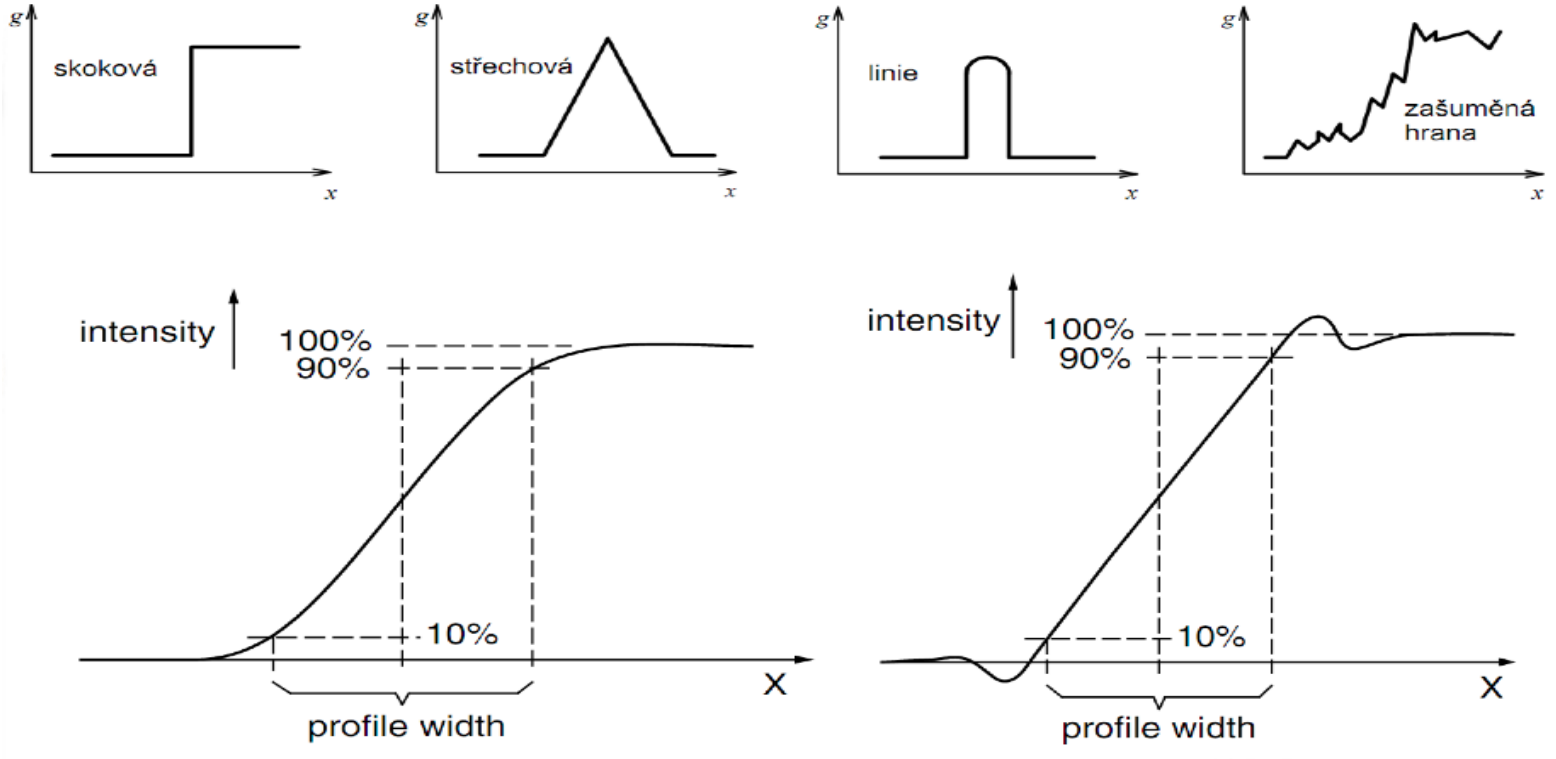
- **Bodové operátory** – pixel ve výstupním obrazu je ovlivněn pouze pixelem na odpovídající pozici ve vstupním obraze

- **Lokální operátory** – pixel výstupního obrazu je ovlivněn pouze omezeným okolím pixelu ve vstupním obrazu (velikost okolí je dána velikostí masky).

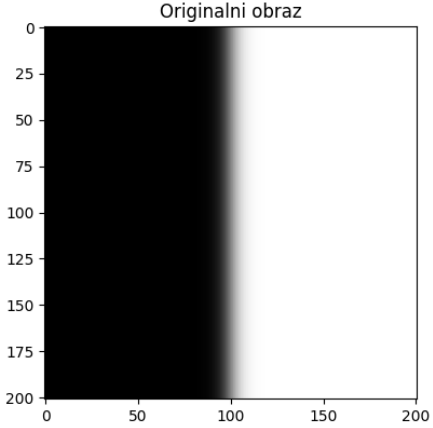
- **Globální operátory** – každý pixel výstupního obrazu je ovlivněn všemi pixely vstupního obrazu.

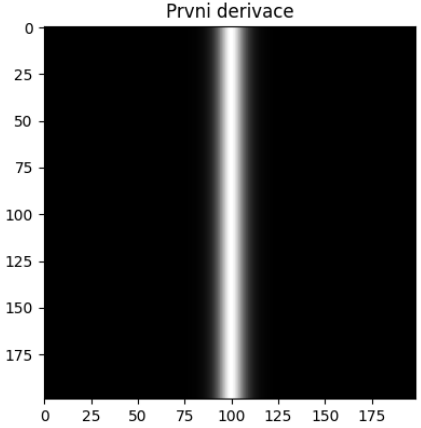


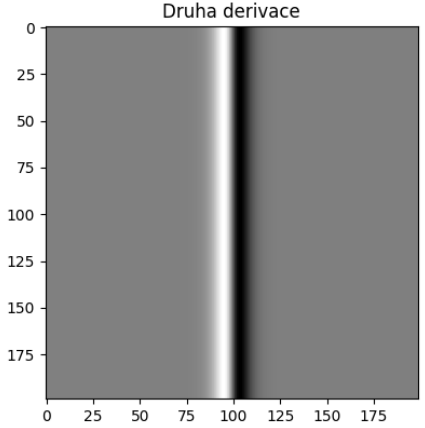
Jasový profil hrany

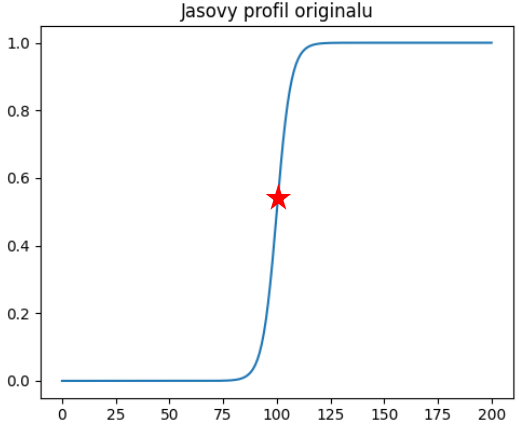


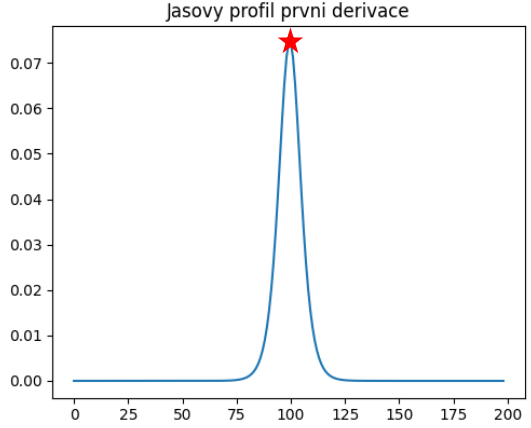
1. a 2. derivace

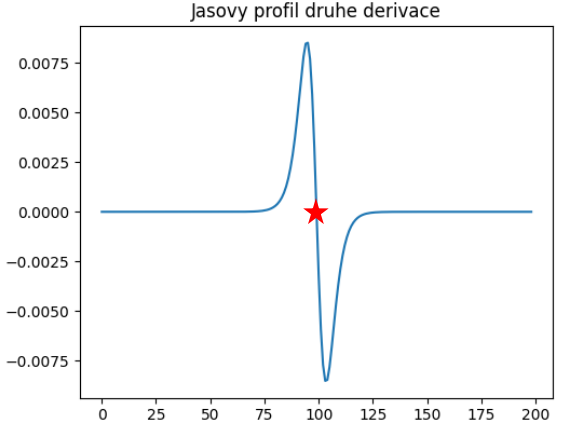












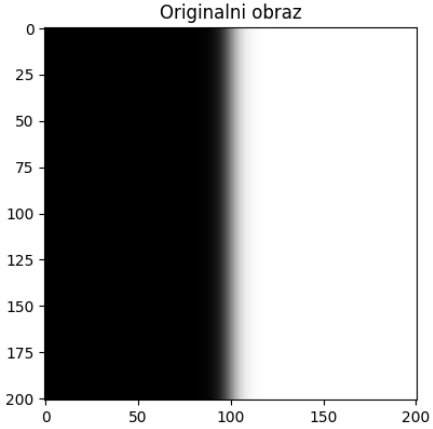
Aproximace 1. a 2. derivace

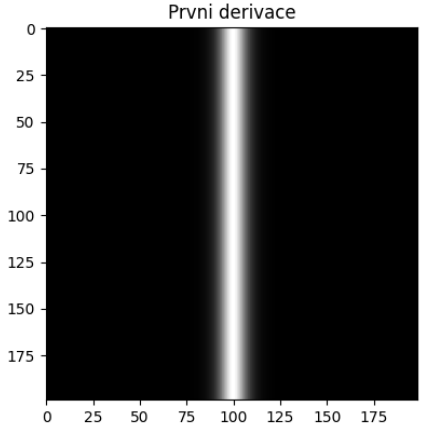
- $\Delta_y f_{i,k}: \begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\Delta_x f_{i,k}: \begin{bmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
- $\Delta_y^2 f_{i,k}: \begin{bmatrix} 0 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ $\Delta_x^2 f_{i,k}: \begin{bmatrix} 0 & 0 & 0 \\ 1 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

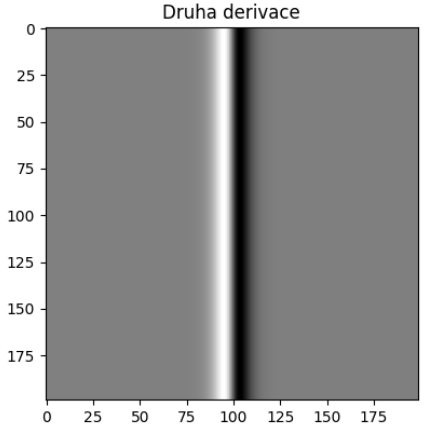
Operátory aproximující 1. derivaci

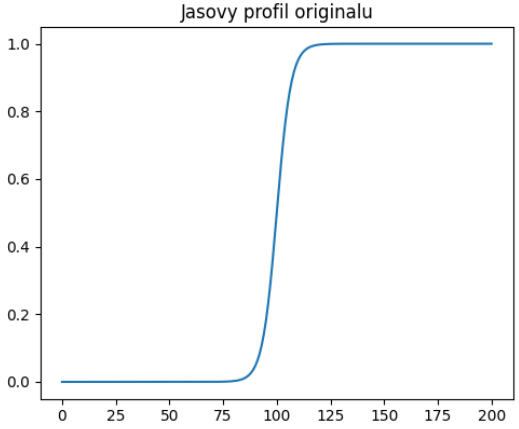
- Robertův operator $h_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $h_2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
- Prewittův operator $h_1 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$ $h_2 = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix}$
- Sobelův operator $h_1 = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$ $h_2 = \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 1 \\ -2 & -1 & 0 \end{bmatrix}$
- Kirschův operator $h_1 = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 0 & 3 \\ -5 & -5 & -5 \end{bmatrix}$ $h_2 = \begin{bmatrix} 3 & 3 & 3 \\ -5 & 0 & 3 \\ -5 & -5 & 3 \end{bmatrix}$

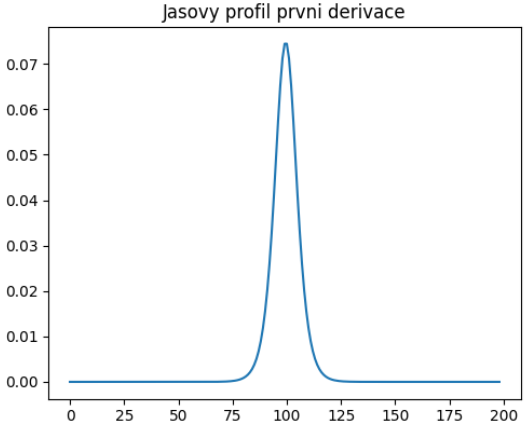
Úkol 1

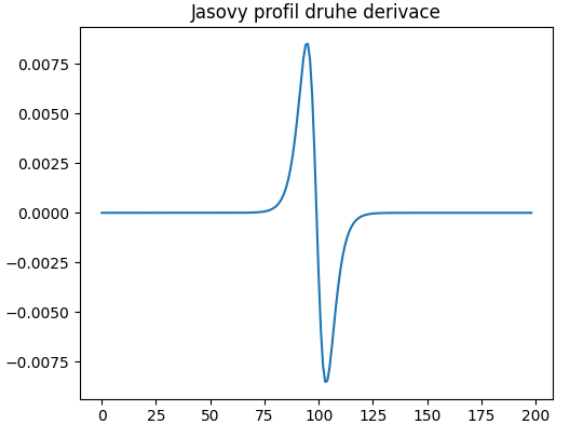

Originalni obraz
0
25
50
75
100
125
150
175
200
0
50
100
150
200


Prvni derivace
0
25
50
75
100
125
150
175
0
25
50
75
100
125
150
175


Druha derivace
0
25
50
75
100
125
150
175
0
25
50
75
100
125
150
175


Jasovy profil originalu
1.0
0.8
0.6
0.4
0.2
0.0
0
25
50
75
100
125
150
175
200


Jasovy profil prvni derivace
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0
25
50
75
100
125
150
175
200


Jasovy profil druhe derivace
0.0075
0.0050
0.0025
0.0000
−0.0025
−0.0050
−0.0075
0
25
50
75
100
125
150
175
200

Operátory aproximující 2. derivaci

- Laplaceův $h_l = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} or \begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ or $\begin{bmatrix} 1 & 2 & 1 \\ 2 & -12 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

- LoG (Laplacian of Gaussian) $h_{LoG} = \begin{bmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & -2 & -1 & 0 \\ -1 & -2 & 16 & -2 & -1 \\ 0 & -1 & -2 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{bmatrix}$

Zostřující operátor

$$\left(\frac{1}{K}\right)\cdot\begin{bmatrix}-1 & -1 & -1\\ -1 & K+8 & -1\\ -1 & -1 & -1\end{bmatrix}=\mathrm{K}\cdot\begin{bmatrix}0 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 0\end{bmatrix}-\begin{bmatrix}1 & 1 & 1\\ 1 & -8 & 1\\ 1 & 1 & 1\end{bmatrix}$$

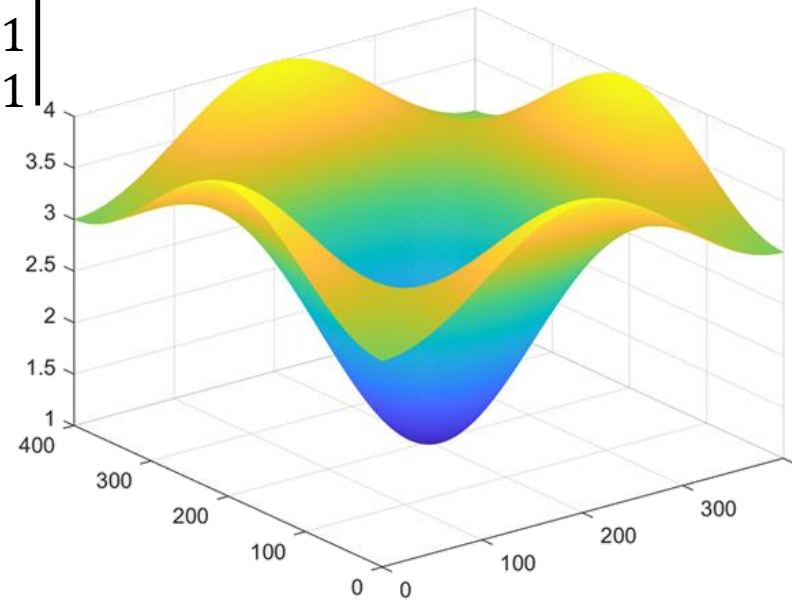
$$\mathbf{G}\qquad\qquad\mathbf{F}\qquad\qquad\mathbf{L}$$

$$\mathrm{G}=\mathrm{K}\cdot\mathrm{F}-\mathrm{L}$$

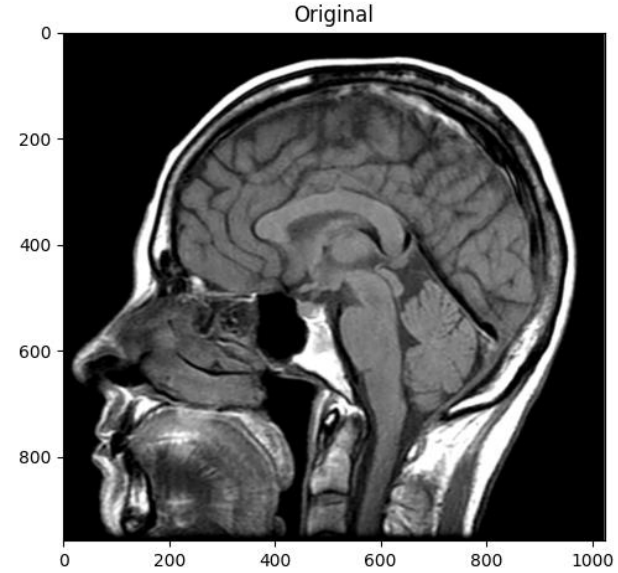
$$\mathrm{G}=\mathrm{F}-\alpha\mathrm{L}$$

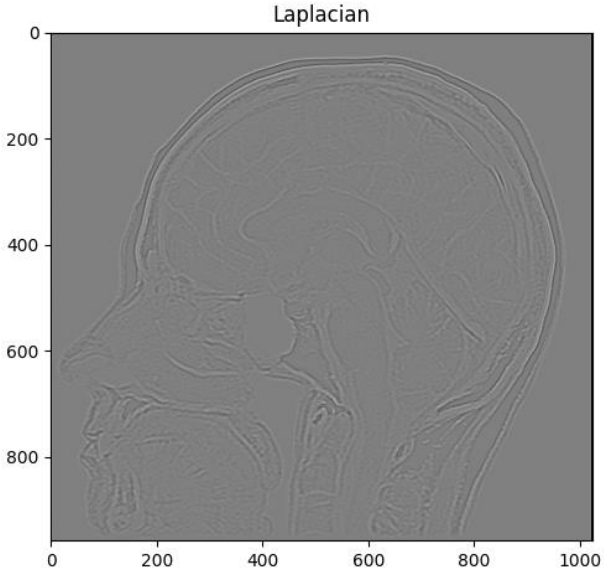
$$\begin{bmatrix}-\alpha & -\alpha & -\alpha\\ -\alpha & 8\alpha+1 & -\alpha\\ -\alpha & -\alpha & -\alpha\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 0\end{bmatrix}-\alpha\begin{bmatrix}1 & 1 & 1\\ 1 & -8 & 1\\ 1 & 1 & 1\end{bmatrix}$$

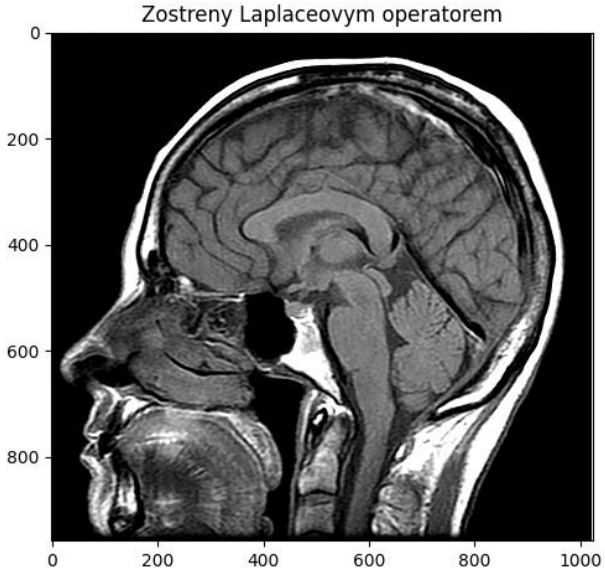
$$\mathbf{G}\qquad\qquad\mathbf{F}\qquad\qquad\mathbf{L}$$



Úkol 2


Original
0
200
400
600
800
0
200
400
600
800
1000


Laplacian
0
200
400
600
800
0
200
400
600
800
1000


Zostreny Laplaceovym operatorem
0
200
400
600
800
0
200
400
600
800
1000

Závěr cvičení

- git add .
- git commit –m „message“
- git push origin main